

# Lecture 7:

AI504 - Knowledge Representation

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# 1 Verbs

We live in a world with 2 types of creatures

- 1) moonacks
- 2) woodchucks

We know all moonacks are woodchucks, but nothing else

- 1) Do all who fear all moonacks fear all woodchucks?
- 2) Do all who fear all woodchucks fear all moonacks?
- 3) Do all who love all who fear all moonacks, love all who fear all woodchucks?
- 4) Do all who love all who fear all woodchucks, love all who fear all moonacks?

– in this example 1 and 3 are unknown, while 2 and 4 must be true.

Verbs are labels as

- 1) Let  $m = \{1\}$ ,  $w = \{1, 2\}$ , where  $3 \xrightarrow{\text{fears}} m$

Then  $\{1\} \subset \emptyset \perp$

- 2) Has to be true

- suppose i fear all woodchucks

Let  $m$  be an arbitrary moonack,  $m$  must be a woodchuck

Therefore I fear  $m$

- Therefore i fear all moonacks

- 3) Let  $m = \{1\}$ ,  $w = \{1, 2\}$ , where  $3 \xrightarrow{\text{fears}} m$  and  $4 \xrightarrow{\text{loves}} 3$

Then  $\{1\} \subset \emptyset \perp$

- 4) Has to be true

- suppose i love an arbitrary person who fears all woodchucks

Let  $m$  be an arbitrary moonack,  $m$  must be a woodchuck

this arbitrary person must fear  $m$

Since I love this person

- I must love all who fears all moonacks

This is confusing 3 should be true, and false should be false